5.3.8.4 *Low-speed external clock generated from a crystal resonator*

The low-speed external (LSE) clock can be supplied with a 32.768 kHz crystal resonator oscillator. All the information given in this paragraph are based on design simulation results obtained with typical external components specified in the table below. In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

Table 34. LSE oscillator characteristics (f_{LSE} = 32.768 kHz)

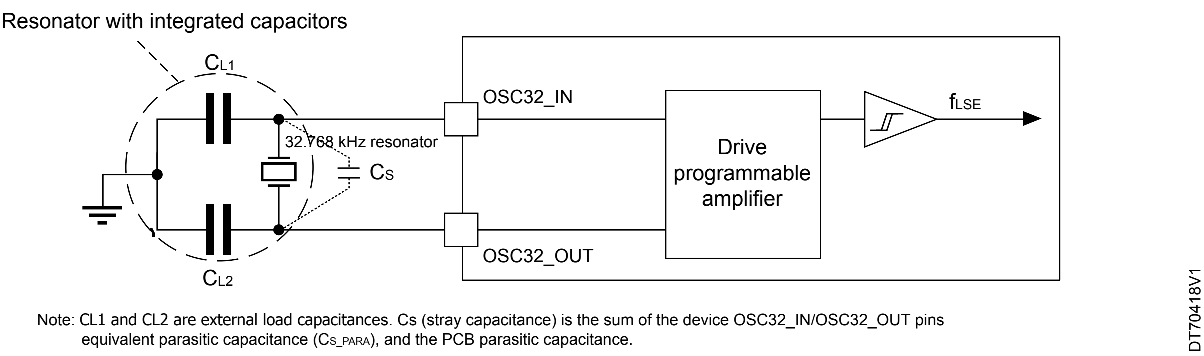
Specified by design and not tested in production.

Symbol	Parameter	Operating conditions(1)	Min	Typ	Max	Unit
F	Oscillator frequency	-	-	32.768	-	kHz
I_{DD}	LSE current consumption	LSEDRV[1:0] = 00, Low drive capability	-	246	-	nA
		LSEDRV[1:0] = 01, Medium low drive capability	-	333	-	
		LSEDRV[1:0] = 10, Medium high drive capability	-	462	-	
		LSEDRV[1:0] = 11, High drive capability	-	747	-	
Gmcritmax	Maximum critical crystal Gm	LSEDRV[1:0] = 00, Low drive capability	-	-	0.5	µA/V
		LSEDRV[1:0] = 01, Medium low drive capability	-	-	0.75	
		LSEDRV[1:0] = 10, Medium high drive capability	-	-	1.7	
		LSEDRV[1:0] = 11, High drive capability	-	-	2.7	
t_{SU}(2)	Startup time	V_{DD} is stabilized	-	2	-	s

1. *Refer to the following note and caution paragraphs, and to the application note AN2867 “Oscillator design guide for ST microcontrollers.*
2. *t_{SU} is the startup time measured from the moment it is enabled (by software) to a stabilized 32.768k Hz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer.*

Note: For information on selecting the crystal, refer to the application note 'Oscillator design guide for STM8AF/AL/S, STM32 MCUs and MPUs' (AN2867).

Figure 18. Typical application with a 32.768 kHz crystal



Note: An external resistor is not required between OSC32_IN and OSC32_OUT and it is forbidden to add one.